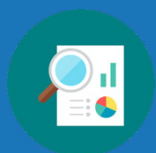


Unit 2 Overview: Fractions, Percentages, and Decimals

Unit Narrative



In this unit, students learn how to rewrite rational numbers in different but equivalent forms, including fractions, decimals, and percentages. Students solve multi-step real-world problems using the four mathematical operations. In the 6th grade accelerated course Unit 1, students focused on performing operations with fractions and decimals fluently, which will apply in this unit when they are solving real-world problems. In the 6th grade accelerated course Units 13 and 14, students will use knowledge learned in this unit to help write expressions and equations to represent real-world contexts, including solving one-step equations that represent those contexts.

Section 1 is focused on the conversion between rational numbers in different but equivalent forms. Lesson 1 provides strategies for converting fractions to decimals and decimals to percentages. Lessons 2 & 3 provide methods and strategies to convert from fractions to percentages. Section 1 emphasizes the importance of different forms being equivalent before transitioning to solving real-world problems involving operations with rational numbers in different forms in Section 2.

Section 2 extends students' understanding of rational number conversions from Section 1 to determining the appropriate mathematical operations to use when solving various real-world problems. Students analyze real-world contexts, including a focus on finance in Lesson 5, to determine which values and operation(s) to use to solve problems.

Section 3 incorporates the skills and strategies learned in Lessons 1-5 to assist students in solving real-world multi-step problems. Lessons 6-7 provide a progression of content from one-step problems to multi-step problems.

The knowledge and skills developed in these first two units of this course will continue to be spiraled throughout the remainder of the course as students develop procedural fluency with rational numbers by the end of the course.

Unit At-A-Glance

Section 1: Equivalent Conversions	Benchmarks Addressed	Lesson 1	Equivalent Conversions: Fractions ↔ Decimals; Decimals ↔ Percentages
	MA.6.NSO.3.5	Lesson 2	Equivalent Conversions: Fractions ↔ Percentages – Part 1
		Lesson 3	Equivalent Conversions: Fractions ↔ Percentages – Part 2
Section 2: Using Operations to Solve Real-World Contexts	Benchmarks Addressed	Lesson 4	Using Operations to Solve Real-World Problems – Part 1
	MA.6.NSO.2.3	Lesson 5	Using Operations to Solve Real-World Problems – Part 2
Section 3: Solving Real-World Problems with Fraction and Decimal Operations	Benchmarks Addressed	Lesson 6	Solving Two-Step, Real-World Problems with Fractions or Decimals
	MA.6.NSO.2.3	Lesson 7	Solving Multi-Step, Real-World Problems with Fractions or Decimals

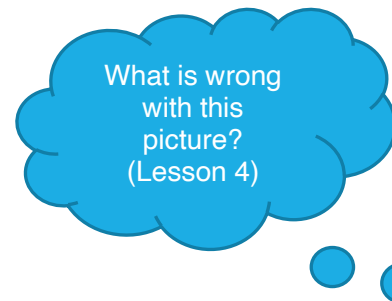


Differentiation

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Engagement The “WHY” of learning

- Lessons 4-5 include contexts about the money. Consider using the content from these lessons as a hook throughout the unit, particularly as students begin engaging with fractions in Lessons 1-3. Ease anxiety with fractions, decimals, and percentages by consistently referencing what students know and are familiar with in everyday life. This content also sets them up for the financial contexts they will encounter in Unit 3 when interpreting positive and negative rational numbers in context.



Representation The “WHAT” of learning

- Provide students with a laminated decimals place-value charts to help them recognize patterns with decimal place value and when multiplying or dividing by powers of 10.
- Utilize 100-grids, area models, and other manipulatives to visually represent converting between equivalent fractions, decimals, and percentages. Connect these representations to equivalent fractions out of 100 in Section 1 to strengthen flexible thinking about equivalent values written as fractions, decimals, and percentages.
- Clarify financial literacy vocabulary in Lesson 5 (Income Finance, Budget, Investment, Principal Return, Gross Income, and Net Income) by providing Frayer Model graphic organizers or other methods for students to write and illustrate examples and nonexamples of these content-specific key words.

Decimal Place Value Chart							
Thousands	Hundreds	Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths

Action & Expression The “HOW” of learning

- Consider how project-based learning in this unit might bring the math of fractions, decimals, and percentages to life for some students. Notice the Warm-Up: Math Flop from Lesson 4 pictured above. Challenge students to go on a scavenger hunt to find similar math flops out in the world, and see what they bring back!
- Provide graph paper or mini coordinate planes with grids for students to align any multiplication or division arithmetic when converting fractions, decimals, and percentages.

ELL Information

Partner Reading can be used to help students approach more complicated and wordy problems as well as opportunities for peer collaboration and community building. This strategy for approaching story problems supports ELL students by providing an opportunity for additional feedback between themselves and their peers in both comprehending the story problem and the corresponding math skills or computations needed to solve. To utilize Partner Reading:

- Prior to the lesson, identify a variety of text-heavy word problems for all operations and one- and two-step problems in a real-world situation within Section 2 and Section 3 of this unit.
- Working in pairs, one student reads the word problem aloud, while the other student assists in pronunciation or assistance when needed.
- When both students have practiced reading and received feedback, they will begin solving the word problem together to find the solution. Students can use a graphic organizer (described below) or other visual mark-up tools to engage with the problem as they read it aloud.
- Whenever possible, act out or provide illustrations of word problems to help students visualize the problem they are going to solve. This may be especially helpful when word problems include new vocabulary, like in Lesson 5's focus on financial literacy and corresponding content-specific vocabulary words.
 - Read problems out loud and gesture or point to the illustration to reinforce the meaning.
 - For some problems, students can draw their own pictures. Try having them do this after you have shown them some examples.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated Support."

- Dividend
- Divisor
- Expression

Additional Differentiated Support

Scaffolding Resources

Graphic Organizers

can be used in Section 3 as a scaffold for students as they identify steps in one- and two-step problems, recognize which operations are needed using the context, and complete arithmetic work to solve real-world problems.

To support these skills, a graphic organizer can help students visually and clearly see what information is needed and the process for approaching the information in the story problems. The easy-to-read format may be beneficial for all learners, as it also helps develop students' abilities to recognize what information is necessary and what information is not needed when solving problems, then organizes that information in a logical process that can be followed as a problem-solving approach.



Consider providing a pre-constructed table for scaffolding practice or encourage students to model their own sensemaking strategies when practicing MTR 7.

Example:

What is the question asking?	Key Words / Phrases ($\pm$, $\times$, $\div$)	My Approach / Strategy

On-Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Prime and Composite Numbers
- Distributive Property
- Multiplying Decimals
- Dividing Decimals
- Multiply a Fraction by a Whole Number
- Multiplying Fractions and Finding Area
- Fractions in Real-World Multiplication Problems
- Understanding Digits to the Right
- Understanding Digits to the Left
- Adding Decimals
- Subtracting Decimals
- Recognize and Generate Equivalent Fractions Using Models
- Interpreting Multiplication Equations
- Real-World Addition/Subtraction Multi-Step Problems
- Real-World Multiplication/Division Multi-Step Problems
- Real-World Multiple Operations Multi-Step Problems



Section Overview

Section 1: Equivalent Conversions (Lessons 1-3)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals and percentages. <i>Clarification 1: Rational numbers include decimal equivalence up to the thousandths place.</i>	Lesson 1	- I can convert between fractions and decimals. - I can convert between percentages and decimals.
	Lesson 2	- I can convert between fractions with denominators of 100 and percentages. - I can convert between fractions and percentages using equivalent fractions.
	Lesson 3	- I can convert between fractions and percentages. - I can rewrite positive rational numbers in different but equivalent forms.

Section 2: Using Operations to Solve Real-World Problems (Lessons 4-5)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers. <i>Clarification 1: Within this benchmark, it is not the expectation to include both decimals and fractions within a single problem.</i>	Lesson 4	I can determine which operation should be used to solve a real-world problem.
	Lesson 5	I can solve real-world problems with up to two steps using decimals or fractions.

Section 3: Solving Real-World Problems with Fraction and Decimal Operations (Lessons 6-7)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers. <i>Clarification 1: Within this benchmark, it is not the expectation to include both decimals and fractions within a single problem.</i>	Lesson 6	I can solve real-world problems with two steps using decimals or fractions.
	Lesson 7	I can solve real-world problems with three steps using decimals or fractions.



Vertical Alignment

Grade 6 Accelerated Unit 2
Fractions, Percentages,
and Decimals



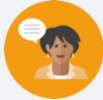
Grade 7 Accelerated Unit 2
Rational and Irrational Numbers

Prior Course(s)	Course Level	Future Course(s)
MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction. MA.5.AR.1.1 Solve multi-step real-world problems involving any combination of the four operations with whole numbers, including problems in which remainders must be interpreted within the context. MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction or multiplication of fractions, including mixed numbers and fractions greater than 1. MA.5.AR.1.3 Solve real-world problems involving division of a unit fraction by a whole number and a whole number by a unit fraction.	MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages. MA.6.NSO.2.3 Solve multi-step real-world problems involving any of the four operations with positive multidigit decimals or positive fractions, including mixed numbers.	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line. MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.

2.1 Equivalent Conversions: Fractions ↔ Decimals; Decimals ↔ Percentages

Lesson Introduction

 Benchmark	MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.		 Learning Targets	- I can convert between fractions and decimals. - I can convert between percentages and decimals.
 Benchmark Coherence	Building On MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.		Working Toward MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.	
 Essential Questions	- How can you use place value to convert a decimal to a fraction? - How do you convert a percentage to a decimal? - What are the similarities and differences between converting fractions, decimals, and percentages?			
 Lesson Narrative	In this lesson, students learn how to convert between fractions and decimals and between percentages and decimals. 2.1.1 Warm-Up activates prior knowledge of place values by analyzing the different numerical values of digits to determine place value. Through Guided Instruction, students make connections between place values in decimals and denominators in fractions as they learn to convert between fractions and decimals, and between decimals and percentages.			
 Component Summary	2.1.1 Warm-Up (~5 min.)	Self-evaluation on place value (SE p. 41)	2.1.4 Guided Instruction (5-10 min.)	Making connections among percentages, decimals, and fractions (SE p. 45)
	2.1.2 Guided Instruction (15-20 min.)	Explore relationships of value, decimals, and fractions (SE p. 42)	2.1.5 Guided Practice (~10 min.)	Practice converting between percentages and decimals (SE p. 45)
	2.1.3 Guided Practice (5-10 min.)	Practice converting between fractions and decimals (SE p. 44)	2.1.6 Your Turn! (~5 min.)	Practice converting between percentages, decimals, and fractions (SE p. 46)
			2.1.7 Wrap-Up (~5 min.)	Which one does not belong activity (SE p. 46)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Percentage <u>Additional Terms:</u> - Place value		(Optional) Place value chart (Optional) Chart paper	
	Additional Preparation (Optional) Create and laminate a decimal place value chart for students to use or reference throughout the lesson and activities. (Optional) Create an anchor chart that can be used in 2.1.2 Guided Instruction			

 Teacher Tips	Common Misconceptions • Students may put the decimal point in the wrong place. • Students may not know where to put the decimal point on a whole number. • Students may mix up the numerator with the denominator. • Students may read decimals incorrectly (0.625 as 625 hundredths instead of 625 thousandths)	Things to Consider • Use student responses from the self-evaluation in 2.1.1 Warm-Up to possibly review the rules for dividing and multiplying decimals and to gauge scaffolding needed for subsequent lessons in this unit related to prior knowledge on fractions and decimals. • When questioning and modeling problems, display and encourage the use of accurate math vocabulary including <i>percentage</i>, <i>decimal</i>, <i>fraction</i>, and <i>place value</i> to help students make connections and notice differences.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 2.1.3 and 2.1.5 Guided Practices provide opportunities for students to explain, justify, agree, or disagree with conversion problems. Consider modeling “Take Turns” routines for the first problems in each section to help students identify and explain the relationship between place value, decimals, fractions, and percentages.	MA.K12.MTR.3.1: Mathematical Fluency Students will be working to convert between decimals, fractions, and percentages. Appropriate and efficient methods are necessary in finding accurate answers and building fluency in recognizing different equivalent forms. Offer students flexibility in converting numbers, while providing opportunities for math-rich discussions about different possible methods.
 Differentiate Instruction	For visual learners, consider providing struggling students with a laminated decimal place value chart to help them when converting with decimals. Consider incorporating a gallery walk for kinesthetic learners.	
Lesson Notes:		

2.1.1 Warm-Up: Self-Evaluation

Component Overview: Students individually rate their confidence on a series of different conversations from decimals, fractions, and/or percentages. Consider using the information provided by this evaluation to assess the amount of scaffolding required in subsequent lessons and throughout this unit.



2.1.1 Warm-Up: Self-Evaluation

- For each problem, rate your level of confidence in your ability to answer each question, but don't actually answer them.

Use a scale from 1 to 5, where 1 means you do not feel confident about how to answer the question, and 5 means you feel very confident.

Problem	Confidence Level
Rewrite $2\frac{2}{5}$ in its equivalent decimal form.	<i>Answers will vary.</i>
Rewrite 0.625 as a fraction with the same value.	<i>Answers will vary.</i>
Rewrite 0.9 as an equivalent percentage.	<i>Answers will vary.</i>
Rewrite 75% in its equivalent decimal form.	<i>Answers will vary.</i>



Consider reviewing the confidence scale results to determine the level of scaffolding, if any, needed during activities in this lesson and subsequent lessons in this unit. Student responses may also be helpful in creating optional groups later in the lesson during Guided Practice sections.

2.1.2 Guided Instruction: Fraction and Decimal Conversions

Component Overview: Students activate prior knowledge of place values by using a decimal place value chart to understand the value of digits in different place values. Students then make connections between place values in decimals and denominators in fractions as they learn to convert between fractions and decimals.



2.1.2 Guided Instruction: Fraction and Decimal Conversions

Decimal Place Value Chart							
Thousands	Hundreds	Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths



Consider displaying or having students use a laminated decimals place value chart to reference place value and decimals. This entry point for visual learners can be used in subsequent activities and lessons in which decimals conversions are practiced.



The place value of a number provides the numerical value that a digit has by virtue of its position in a number.

Example: In 278.493

the 8 is in the **ones** place

the 4 is in the **tenths** place

the 9 is in the **hundredths** place

the 3 is in the **thousandths** place

1. Use the original number to fill in the missing digits and place values.

Original Number	Digit	Place Value
422.54	5	Tenths
5.276	5	Ones
12.749	4	<i>Hundredths</i>
3.826	6	Thousandths

2.1.2 Guided Instruction: Fraction and Decimal Conversions (continued)

2. Consider the number 0.08.
- In the number 0.08, what is the place value of the final digit?
Hundredths
 - What is the written description of the fraction $\frac{8}{100}$?
8 hundredths
 - Discuss with your partner any similarities or differences between the decimal 0.08 and the fraction $\frac{8}{100}$.



When rewriting a decimal that is greater than 0 but less than 1 as an equivalent fraction, use the place value of the final digit to determine the denominator of the fraction.

3. In the decimal 0.7, the final digit is in the

- ☐ tenths
☐ hundredths
☐ thousandths

place, so

☐ $\frac{7}{10}$
☐ $\frac{7}{100}$
☐ $\frac{7}{1000}$

is the equivalent

fraction form.

4. In the decimal 0.037, the final digit is in the

- ☐ tenths
☐ hundredths
☐ thousandths

place, so

☐ $\frac{37}{10}$
☐ $\frac{37}{100}$
☐ $\frac{37}{1000}$

is the equivalent

fraction form.



Notice the question prompts in questions 2a-c. Position students to discover the relationship between the place value and the fraction equivalent by exploring questions 2a-c with a partner.



Monitor student responses and listen for opportunities to extend their thinking.

- “The decimal and fraction are the same.” What do you mean by the same? Which part is the same? What do we call that in math? (equivalent)
- “They both have two zeros.” Will that always be true? Which part of the expression determines ‘how many zeros’ or the place value of the fraction? Does that work for this example?



Are your students ready for a challenge? Consider briefly incorporating a number like 5.45 or 109.032 to see what students do with the 5 and the 109. Some may omit the values, while others may place them in the numerator, while others may represent the expression as a mixed number.

2.1.2 Guided Instruction: Fraction and Decimal Conversions (continued)

A fraction can be converted into its equivalent decimal form by dividing the **numerator** by the **denominator**.

5. To convert the fraction $\frac{1}{4}$ to a decimal, divide 1 by 4 by completing the diagram below.

		0	.	2	5
	4	)	1	.00	
Number of ones groups of 4	—		0		
Amount that remains			10		
Number of tenths groups of 4	—		8		
Amount that remains			20		
Number of hundredths groups of 4	—		20		
Amount that remains			0		



Consider an anchor chart for long division that highlights conceptual understanding while supporting a variety of entry points.



Why do we need to write the decimal to the right of the whole number dividend in this problem? Why is it important to write the decimal in the quotient lined up with the same place of the dividend?

2.1.3 Guided Practice: Fraction and Decimal Conversions

Component Overview: Students convert between equivalent fraction and decimal forms.

2.1.3 Guided Practice: Fraction and Decimal Conversions

- Complete the table by converting each decimal to an equivalent fraction. If possible, write the equivalent fraction so that the numerator and denominator have no common factors.

Decimal Form	Fraction Form
0.08	$\frac{2}{25}$
0.27	$\frac{27}{100}$
2.2	$2\frac{2}{10} = 2\frac{1}{5}$
0.005	$\frac{5}{1000} = \frac{1}{200}$

- Complete the table by rewriting each fraction as its equivalent decimal.

Fraction Form	Your Work	Decimal Form
$\frac{1}{5}$	Answers will vary.	0.2
$\frac{5}{8}$	Answers will vary.	0.625
$\frac{7}{200}$	Answers will vary.	0.035



Take Turns

Use Take Turns language routines for question 1 by providing opportunities for students to explain, justify, agree, or disagree with answers given by peers. Support student discussions with sentence stems, including:

- I know ___ is equivalent to ___ because...
- I disagree because...
- I agree because...



Monitor student responses while they are practicing. Common misconceptions could include:

- Misplacing the decimal (2.0 instead of 0.2)
- Mixing up the numerator and denominator
- Reading decimals (0.625 as 625 hundredths instead of 625 thousandths)

2.1.4 Guided Instruction: Decimal and Percentage Conversions

Component Overview: Students use knowledge of fractions and decimals to determine the connection to percentages. The definition and notation of percentages are introduced, and students learn how to convert between decimals and percentages.



2.1.4 Guided Instruction: Decimal and Percentage Conversions

A **percentage** is a ratio expressed as a fraction with a denominator of 100.

Percentages can also be written as fractions with a denominator of 100, or as a decimal.

- Write 66% as a fraction and as a decimal.

Percent	Fraction	Decimal
66%	$\frac{66}{100}$	0.66

To convert a decimal to a percent, multiply the decimal by 100.

- Convert the decimal 0.16 to a percent.

$$0.16 \times \frac{100}{100} = 16\%$$

To convert a percent to a decimal, divide the percent by 100 or multiply by $\frac{1}{100}$.

- Convert 16.2% to a decimal.

$$16.2 \div \frac{100}{100} = 0.162 \text{ or } 16.2 \cdot \frac{1}{100} = 0.162$$



To strengthen student understanding of percentage as a ratio out of 100, ask students to think of words they know that contain “cent,” meaning 100, then have them explain how the word relates to 100. Examples include:

- A cent = $1/100^{\text{th}}$ of a dollar; 100 cents in a dollar
- Century = 100 years
- Centimeter = $1/100^{\text{th}}$ of a meter; 100 centimeters in a meter
- Centipede = a bug with 100 legs

2.1.5 Guided Practice: Decimal and Percentage Conversions

Component Overview: Students convert between different equivalent decimal and percentage forms.



2.1.5 Guided Practice: Decimal and Percentage Conversions

- Complete the table by converting each decimal to an equivalent percentage.

Decimal Form	Percent Form
0.08	8%
0.27	27%
2.2	220%
0.005	0.5%



For question 1, what do you notice about the percentage conversions for 2.2 and 0.005? Why is 2.2 written as a percent larger than 100? Why is 0.005 written as a percent smaller than 1%, even though the number 5 is greater than the number 2?

- Rewrite each percentage into its equivalent decimal form to complete the table.

Percent Form	Decimal Form
72%	0.72
12.5%	0.125
0.7%	0.007
214%	2.14



Consider having students engage in Take Turns language routines by verbally explaining how they converted the percentages to decimal form in question 2. This can position students to explain, justify, agree, or disagree with answers provided by peers.

2.1.6 Your Turn!

Component Overview: Students practice converting between decimals, fractions, and percentages with multiple problems. Consider having students complete some or all the problems based on available time and necessary practice for the different representations.



2.1.6 Your Turn!

1. Rewrite each percentage as a decimal.

a. 125.6% **1.256** b. 3% **0.03**

2. Convert each decimal to an equivalent percentage.

a. 0.008 **0.8%** b. 4.71 **471%**

3. Rewrite each fraction as a decimal.

a. $\frac{1}{2}$ **0.5** b. $2\frac{1}{10}$ **2.1**

4. Rewrite each decimal as an equivalent fraction. Simplify where applicable.

a. 0.94 $\frac{94}{100} = \frac{47}{50}$ b. 1.7 $1\frac{7}{10}$



One of the most common misconceptions finds students writing single digit percentages as tenths, like question 1b students writing 3% as 0.3. Consider asking probing questions that position students to connect percent as “out of 100” in order to conceptualize $3\% = \frac{3}{100} = \text{three hundredths} = 0.03$.



Consider using graph paper or base-10 blocks as visual scaffolding for aligning and categorizing place values and decimals.

2.1.7 Wrap-Up: Ticket Out the Door

Component Overview: This Wrap-Up assesses students' ability to convert between fractions, decimals, and percentages and determine which expressions are equal by identifying which representation does not belong. Consider providing written feedback to each student on their explanation and include one point of positive feedback to the student.



2.1.7 Wrap-Up: Ticket Out the Door

1. Circle the expression that does not belong.

$\frac{2}{5}$	0.4
40%	$\frac{4}{8}$

2. Explain your reasoning for your choice.

Answers will vary. $\frac{4}{8}$ is equivalent to $\frac{1}{2}$ which is written as 0.5 as a decimal and 50% as a percentage. $\frac{2}{5}$, 0.4, and 40% are all equivalent.



Consider how differing responses can tell you different elements of students' learning progression.

- Based on the learning targets for this lesson, most students might choose $\frac{4}{8}$ to correctly identify as the only expression not equivalent.
- Students may also choose the decimal or percent as the only decimal or percent shown, while other students may identify $\frac{2}{5}$ as the only option without a "4" in it, which would be a good opportunity to probe students on equivalence and multiple representations.
- A "correct" response should not be the focus as much as the student using appropriate mathematics language in their reasoning.

2.2 Equivalent Conversions: Fractions ↔ Percentages – Part 1

Lesson Introduction

 Benchmark	MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.		 Learning Targets	- I can convert between fractions with denominators of 100 and percentages. - I can convert between fractions and percentages using equivalent fractions.
 Benchmark Coherence	Building On MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.		Working Toward MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.	
 Essential Questions	- What is a percentage? - How do percentages relate to fractions? - How do you convert fractions to percentages and percentages to fractions?			
 Lesson Narrative	In this lesson, students learn how to convert between fractions with denominators of 100 and percentages, and how to convert between fractions and percentages using equivalent fractions. Students begin with a video on the history of the numerical system, then explore the meaning of percentage and how it relates to fractions with a denominator of 100. Students will then use this newly acquired knowledge to form equivalent fractions to percentages using multiplication and equivalent fractions.			
 Component Summary	2.2.1 Warm-Up (~5 min.)	Place Value System Video (SE p. 47)	2.2.3 Guided Practice (10-15 min.)	Practice converting fractions to percentages using equivalent fractions (SE p. 49)
	2.2.2 Guided Instruction (10-15 min.)	Exploration on percentages and relationship to fractions with a denominator of 100 (SE p. 48)	2.2.4 Your Turn! (10-15 min.)	Students practice forming equivalent values and converting fractions to percentages using equivalent fractions (SE p. 50)
	2.2.5 Wrap-Up (~5 minutes)	Create equivalent fractions (SE p. 51)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Percent <u>Additional Terms:</u> - Equivalent fractions - Factors		- Numerical Systems Video (Optional) Chart paper (Optional) Markers (Optional) 100-grids	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may believe a reduced fraction to be 'new' or 'different' than the original (or equivalent). - Students may only perform the multiplication operation on the denominator to get it equal to 100. - Students may interpret single digit percentages as tenths place as decimals (i.e., 2% = 0.2).	- Highlight the equivalence of representations of fractions or decimals ($\frac{1}{2}$ is the same value as $\frac{5}{10}$). - Consider offering factor trees and factor cards for students to find factors quickly, spending more time with the concept of equivalence instead of just on factoring.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 2.2.2 Guided Instruction and 2.2.3 Guided Practice position students to identify, compare, and contrast mathematical representations of fractions and percentages, and methods for conversion. Purposefully using words like "compare" or "connect" in discussion prompts students to make connections between prior knowledge and new concepts.	MA.K12.MTR.2.1: Multiple Representations 2.2.2 Guided Instruction includes multiple entry points in representing fractions and percentages in different forms. Consider highlighting these representations and encouraging students to make connections or build their own models with manipulatives or other visual methods, showing how percentages and fractions can represent an equivalent value.
 Differentiate Instruction	Student choice, small-group learning, and leveling problems are all strategies to differentiate content for all students to engage. Consider one of these strategies for implementing Your Turn! For visual learners, consider providing students with graph paper, 100-grids, or math manipulatives to visually represent converting between equivalent fractions and percentages in 2.2.2 Guided Instruction and 2.2.3 Guided Practice. This may also activate prior knowledge of fractions from previous grades. Have students create an anchor chart using the problems from 2.2.4 Your Turn! on how to convert percentages and fractions instead of completing all the problems in this section. This activity may engage visual learners by helping them represent their understanding on paper while also engaging students with a choice in how they design their poster.	

Lesson Notes:

2.2.1 Warm-Up: A Brief History of Numerical Systems

Component Overview: Play the “A Brief History of Numerical Systems” video about how the place value system was perfected by Indian mathematicians in the 8th century. After students watch this 5-minute video, prompt the students to read and answer the questions in the Warm-Up independently.



2.2.1 Warm-Up: A Brief History of Numerical Systems

Humans went from using body parts, tally marks, and symbols to count objects, to what we now know as the base-ten system. Multiple civilizations around the world developed positional numbering systems, and both Mayan and Indian civilizations developed the concept of “zero.” The base-ten system was perfected by eighth century Indian mathematicians. They realized that they could combine the concept of “zero” with a positional system that used repeatable symbols, whose values are determined by their position in a sequence. A base of ten was selected because, quite simply, we have ten fingers. Centuries later, this Hindu-Arabic numeral system began to spread to Europe and replaced the Roman numeral system. This is what we use today.



Consider an extension project featuring the contribution of various cultures to the number system.

- India and the Base-10 System
- The Mayans and Math
- Mathematicians of the African Diaspora

1. What is the historical importance of the number zero?

Answers will vary. Zero helped form the base-ten system. Zero brought many cultures ideas about the number system together.

2. If zero was never created, how would that affect how we do math today?

Answers will vary. Without zero, the base-ten system would not exist. Without zero, we would still use Roman Numerals.

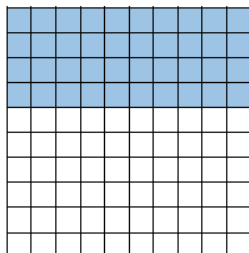
2.2.2 Guided Instruction: Converting Between Fractions and Percentages

Component Overview: This section guides students to understand the meaning of percentage and how it relates to fraction with a denominator of 100.



2.2.2 Guided Instruction: Converting Between Fractions and Percentages

1. The model represents a numerical value. Use the model to answer the questions that follow.



- a. How many total squares are in the image?

100

- b. How many of the squares are shaded blue?

40

- c. What numerical value does the model represent, expressed as a fraction of $\frac{\text{blue shaded squares}}{\text{total squares}}$?

$$\frac{40}{100} = \frac{4}{10} = \frac{2}{5}$$

Recall that a **percentage** is a ratio expressed as a fraction of 100.

Percent (%) means “per 100,” “for every 100,” or “out of 100.”

For example, 40% means “40 out of 100,” so 40% is equivalent to the fraction $\frac{40}{100}$.

2. What decimal value has the same value as 40% and $\frac{40}{100}$?

Equivalent Values		
Percent Form	Fraction Form	Decimal Form
40%	$\frac{40}{100}$	0.40



For visual learners, consider having a 100-grid paper printed and laminated (or multiple 100-grids on one paper) for students to practice shading in different percentages throughout this lesson. This entry point may help students make the connection by coloring in the value out of 100 that is being represented by the percentage.



Utilize Lesson 1 resources (definitions and video) for additional support in “percent” and “percentage” out of 100.

2.2.2 Guided Instruction: Converting Between Fractions and Percentages (continued)

3. With your partner, complete the table by rewriting each value in the equivalent form indicated.

Fraction Form	Percent Form
$\frac{50}{100}$	50%
$\frac{71}{100}$	71%
$\frac{2}{100}$	2%
$\frac{150}{100}$	150%
$\frac{42}{100}$	42%



Compare and Connect

Position students to make connections between the “out of 100” forms in question 3 and original fractions in question 4.

- How is $\frac{6}{10}$ the same? How is it different?
- Why would $\frac{6}{10}$ not be 6%?
- What does that make you think about when converting fractions to percentages?

When converting fractions to percentages, generate an equivalent fraction with a denominator of 100 when possible.

4. Complete the table.

Original Fraction	Equivalent Fraction with a Denominator of 100	Percentage
$\frac{6}{10}$	$\frac{6}{10} \times \frac{10}{10} = \frac{60}{100}$	60%
$\frac{23}{50}$	$\frac{23}{50} \times \frac{2}{2} = \frac{46}{100}$	46%
$\frac{1}{5}$	$\frac{1}{5} \times \frac{20}{20} = \frac{20}{100}$	20%

2.2.3 Guided Practice: Converting Between Fractions and Percentages

Component Overview: In small groups, students convert between fractions with denominators and percentages, and convert between fractions and percentages using equivalent fractions.



2.2.3 Guided Practice: Converting Between Fractions and Percentages

- Complete the table by rewriting each value in the equivalent form indicated. Simplify where applicable.

Fraction Form	Percent Form
$\frac{20}{50}$	40%
$\frac{13}{25}$	52%
$\frac{9}{10}$	90%
$\frac{30}{100} = \frac{3}{10}$	30%
$\frac{18}{100} = \frac{9}{50}$	18%
$\frac{5}{4}$	125%
$\frac{200}{100} = \frac{2}{1} = 2$	200%



Notice the first three rows compared to rows four and five. If students are stuck or unsure how to convert the percentage to a fraction, prompt with questioning:

- What do we know about percentages? (out of 100)
- What would be “out of 100” here?
- What does that look like?
- Can this be reduced? How?

2.2.4 Your Turn!

Component Overview: Students convert between fractions and percentages and convert between percentages and fractions using equivalent fractions. Consider choosing specific questions based on learners' needs or having students use these problems to create their own anchor chart to summarize the processes and methods for converting fractions and decimals.



2.2.4 Your Turn!

1. Rewrite each fraction as an equivalent percentage.

a. $\frac{17}{100}$ 17% b. $\frac{42}{50}$ 84% c. $\frac{36}{25}$ 144% d. $\frac{96}{100}$ 96%

e. $1\frac{9}{100}$ 109% f. $\frac{24}{20}$ 120% g. $\frac{1}{50}$ 2% h. $\frac{4}{5}$ 80%

2. Convert each to a fraction. Simplify where applicable.

a. 90% $\frac{9}{10}$ b. 167% $1\frac{67}{100}$ c. 86% $\frac{43}{50}$ d. 33% $\frac{33}{100}$

e. 226% $2\frac{13}{50}$ f. 29% $\frac{29}{100}$ g. 50% $\frac{1}{2}$ h. 18% $\frac{9}{50}$



Consider strategically differentiating during Your Turn! to increase entry points for students.

- **Choice:** Increase student agency by allowing students to choose four fractions from question 1 and 4 percentages from question 2. Then they must find a classmate who completed each question and exchange check marks for correct answers until all have been verified.
- **Small Group:** Pull a small group based on observational data from the lesson for intervention while other students are practicing independently.
- **Levels:** Identify problems with low or high entry points and differentiate between “levels” in which students must “graduate” from level to level (i.e., $\frac{17}{100}$ would be Level 1, $\frac{42}{50}$ would be Level 2, and $\frac{24}{20}$ would be Level 3).

2.2.5 Wrap-Up: Ticket Out the Door

Component Overview: Students use background knowledge on equivalent fractions and newly acquired knowledge of percentages to write three additional equivalent fractions to a given percentage. This open-ended Wrap-Up assesses students' ability to convert between fractions and percentages using equivalent fractions.



2.2.5 Wrap-Up: Ticket Out the Door

- Use the numbers below to create three more fractions equivalent to any three percentages in the table.

15	4	21	30	25	40
32	1	8	2	6	10
7	50	72	5	80	3



For an added challenge, consider having students write as many equivalent fractions as they can think of (given a time limit). Consider adding a recognition or shout-out during the Warm-Up for the next lesson to the student who generated the most equivalent expressions.

10%	20%	30%	40%	50%	60%	70%	80%	90%
	$\frac{1}{5}$			$\frac{1}{2}$				
	$\frac{2}{10}$			$\frac{2}{4}$				
$\frac{1}{10}$	$\frac{3}{15}$		$\frac{2}{5}$	$\frac{3}{6}$	$\frac{3}{5}$		$\frac{4}{5}$	
$\frac{3}{30}$	$\frac{5}{25}$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{4}{8}$	$\frac{6}{10}$	$\frac{7}{10}$	$\frac{8}{10}$	
$\frac{4}{40}$	$\frac{6}{30}$	$\frac{15}{50}$	$\frac{6}{15}$	$\frac{5}{10}$	$\frac{15}{25}$	$\frac{21}{30}$	$\frac{32}{40}$	$\frac{72}{80}$
$\frac{5}{50}$	$\frac{8}{40}$		$\frac{10}{25}$	$\frac{15}{30}$	$\frac{30}{50}$		$\frac{40}{50}$	
	$\frac{10}{50}$			$\frac{25}{50}$				

Answers will vary.

2.3 Equivalent Conversions: Fractions ↔ Percentages – Part 2

Lesson Introduction

 Benchmark	MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.		 Learning Targets	- I can convert between fractions and percentages. - I can rewrite positive rational numbers in different but equivalent forms.
 Benchmark Coherence	Building On MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.		Working Toward MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.	
 Essential Questions	- What strategies can be used to convert fractions and decimals? - What strategies can be used to convert fractions and percentages? - Why is it helpful to convert a fraction into a decimal first when converting to percentages?			
 Lesson Narrative	In this lesson, students continue to convert between fractions and percentages, now focusing on rewriting positive rational numbers in all three different but equivalent rational forms: decimals, fractions, and percentages. The Guided Instruction positions students to develop fluency when rewriting positive rational numbers in different but equivalent rational forms, by recognizing when it is helpful to first convert fractions to decimals if the denominator is not a factor of 100. Students then work in partners to develop and discuss strategies for rewriting rational numbers, and practice computational accuracy.			
 Component Summary	2.3.1 Warm-Up (~5 min.)	Matching decimals with equivalent percentages puzzle (<i>SE p. 52</i>)	2.3.3 Collaboration (5-10 min.)	Word Bank for formalizing strategies when converting between fractions and percentages (<i>SE p. 54</i>)
	2.3.2 Guided Instruction (15-20 min.)	Guided review on how to convert between decimals, fractions, and percentages (<i>SE p. 52</i>)	2.3.4 Guided Practice (5-10 min.)	Individual practice rewriting rational numbers in equivalent forms (<i>SE p. 54</i>)
	2.3.5 Wrap-Up (5 min.)	Assessment of learning targets rewriting rational numbers in equivalent forms (<i>SE p. 55</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Percent		(Optional) Chart paper (Optional) Markers	If blank versions of the tables are being used as suggested in Differentiate Instruction, prepare these tables on chart paper prior to the lesson.

 Teacher Tips	Common Misconceptions - Students may have trouble remembering to divide the numerator by the denominator when converting fractions to decimals. - Students may not recognize equivalent fractions. - Students may confuse when to multiply or divide by 100 when converting between decimals and percentages.	Things to Consider - Consider emphasizing equivalent fractions and strategies for finding equivalent fractions at the start of the lesson. - Encourage students to use multiplication and division reasoning as they write equivalent fractions. - Consider further building students' fluency with factors and the greatest common factor and least common multiple. - If time is a constraint, assign students 2.3.4 Guided Practice as homework.
	Literacy and Language Routines Compare and Connect 2.3.2 Guided Instruction provides students with an opportunity to compare converting fractions, decimals, and percentages. Students can discuss the similarities and differences in calculating the equivalent forms in all three tables in this activity. Consider writing student strategies in a shared classroom space for consideration in 2.3.3 Collaboration.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection Throughout this lesson, students engage in discussions that reflect on their own mathematical thinking and others' thinking in relation to strategies for rewriting rational numbers in equivalent forms. In 2.3.3 Collaboration, students utilize mathematical vocabulary while working together to analyze and communicate methods needed to complete the incomplete strategy statements.
	 Differentiate Instruction	
As an entry point for kinesthetic learners, blank versions of the tables in 2.3.2 Guided Instruction and word bank activities in 2.3.3 Collaboration can be displayed on chart to provide the opportunity for learners to move around the room while working, or work in jigsaw groups to finish one table and present their learning to another. Consider using observations of students' understanding from 2.3.2 Guided Instruction to pull students for small-group instruction for 2.3.4 Guided Practice. Consider displaying the five strategy statements from 2.3.3 Collaboration in a shared classroom space to anchor the various strategies for visual learners.		
Lesson Notes:		

2.3.1 Warm-Up: Matching Puzzle

Component Overview: In this activity, ask students to individually attempt to complete the puzzle by matching decimals to their equivalent percentages, then check their work with a partner. Then have students share the strategies they used to determine the decimal and percentage equivalences.



2.3.1 Warm-Up: Matching Puzzle

Several values, expressed as decimals and percentages, are given.

0.71	710%
7.1	7.1%
0.071	71%

- Match the values in the left column with their equivalent form in the right column by drawing lines to connect them.

See above.

- Describe how you determined which values are equivalent.

Answers may vary. I multiplied each value in the left table by 100 and matched it to the corresponding value in the right table.



Consider having students verbally describe how they determined the equivalent percentage. Highlight strategies that use place value and the connection between percentages representing values “per 100.”

Questioning to address misconceptions related to converting decimals and percentages might include:

- How does the decimal form you connected to the percentage relate to a value out of 100?
- What do you notice about how the decimal placement in the decimal compares to the percent form? How does this connect to a strategy for converting decimals to percentages?
- Would you divide or multiply by 100 to rewrite decimals as percentages? Why?

2.3.2 Guided Instruction: Rewriting Rational Numbers in Equivalent Forms

Component Overview: Students work individually or in small groups to convert between decimals, fractions, and percentages. Consider guiding students through the first table in question 1 in this section as a whole group to review how to divide fractions to convert to decimals, then students can work independently or in small groups to complete the problems in the rest of this section. After finishing the tables, arrange students in small groups to discuss methods and strategies used in the conversions.



2.3.2 Guided Instruction: Rewriting Rational Numbers in Equivalent Forms

Recall from Lesson 1 that a fraction can be converted into its equivalent decimal form by dividing the **numerator** by the **denominator**.

- Complete the table by rewriting the fraction into an equivalent decimal form.

Fraction Form	Decimal Form
$\frac{13}{40}$	0.325
$\frac{7}{8}$	0.875
$2\frac{6}{10}$	2.6

- Complete the table by rewriting the decimal into an equivalent percentage.

Decimal Form	Percent Form
0.325	32.5%
0.875	87.5%
2.6	260%

- What is the equivalent decimal form of $\frac{5}{8}$?
0.625



Compare and Connect

After students complete the tables in questions 1 and 2, consider having students share strategies in small groups to compare the processes before sharing whole group.

Consider writing these strategies on a shared classroom space for students to reference throughout the lesson and to connect to the strategy statements in 2.3.3 Collaboration.



To encourage mathematical reasoning as students complete the tables, ask questions like:

- How do you know those forms are equivalent?
- For the table in question 1, how was your strategy different when converting $\frac{13}{40}$ and $\frac{7}{8}$? Did you use equivalent fractions or division? Why?
- Should a number less than one, in fraction form, also have an equivalent decimal form that is less than 1? What about from decimal to percent form?

2.3.2 Guided Instruction: Rewriting Rational Numbers in Equivalent Forms (continued)

4. What is 0.625 rewritten as a percentage?

62.5%

5. What is $\frac{5}{8}$ rewritten as a percentage?

62.5%

6. When rewriting a fraction as a percentage, why might it be helpful to first rewrite it in its equivalent decimal form?

Answers will vary. Rewriting a fraction as its equivalent decimal can make converting the fraction to a percentage easier because the decimal form only requires you to multiply by 100, or move the decimal place two places to the right.

7. Complete the table by rewriting each value in its equivalent forms.

Fraction Form	Decimal Form	Percent Form
$\frac{675}{1000} = \frac{27}{40}$	0.675	67.5%
$\frac{19}{1000}$	0.019	1.9%
$\frac{63}{20}$	3.15	315%



Compare and Connect

Consider giving students an opportunity to share with a partner or whole group, targeting benchmark-aligned concepts.

- Connection between powers of 10 and place value, including percentage as “out of 100.”
- Purpose of multiplying or dividing by 100, or the connection to “moving” the decimal point.



Using observations on student understanding in 2.3.2 Guided Instruction, consider using the table in question 7 as an opportunity for a teacher-facilitated small group for learners who need additional support. This small group can take place while other students work on 2.3.3 Collaboration.

2.3.3 Collaboration: Strategies for Rewriting Rational Numbers

Component Overview: Students use a word bank to complete five statements describing different strategies for rewriting rational numbers in equivalent form. Additionally, consider having students write these statements and corresponding examples in a math journal or notebook to reference in upcoming lessons.



2.3.3 Collaboration: Strategies for Rewriting Rational Numbers

Working with your partner, use the words in the bank to complete each statement. Words can be used more than once.

Word Bank

decimal	denominator	fraction
numerator	percentage	place value

1. To convert a *fraction* to a decimal, divide the *numerator* by the *denominator*.
2. Multiply the *decimal* by 100 to convert a *decimal* to a *percentage*.
3. To rewrite a *percentage* as a *decimal*, divide the *percentage* by 100.
4. Use the *place value* of the final digit to determine the denominator of the decimal when rewriting a *decimal* as an equivalent *fraction*.
5. An equivalent fraction with a *denominator* of 100 can be used to convert a *fraction* to a percentage.



Consider opportunities to extend this activity for a variety of learning environments.

- **Roles:** Each partner has a role; one student only reads the statements and the other matches the correct word (pair strategically).
- **Charts:** Display each statement on chart paper around the room and students have slips of paper with each word to tape or glue onto each poster.
- **Matches:** Students are given different words or statements and must walk around the room to find their “match.”



If students get stuck, consider prompting them to recognize or apply certain examples or strategies they utilized in previous activities to statements they are reading.

2.3.4 Guided Practice: Rewriting Rational Numbers in Equivalent Forms

Component Overview: Students practice converting between different forms of rational numbers by completing the missing components of a table. The table includes a mix of different forms to assess students' ability to move from any given form to another.



2.3.4 Guided Practice: Rewriting Rational Numbers in Equivalent Forms

1. Complete the table by rewriting each value in its equivalent forms.

Fraction Form	Decimal Form	Percent Form
$\frac{3}{8}$	0.375	37.5%
$\frac{22}{100} = \frac{11}{50}$	0.22	22%
$\frac{4}{100} = \frac{1}{25}$	0.04	4%
$2\frac{33}{50}$	2.66	266%



Compare and Connect

Consider opportunities for students to connect the strategy they used in these conversions with their statements in the previous activity. This will formalize their learning and processes.

2.3.5 Wrap-Up: Ticket Out the Door

Component Overview: This Wrap-Up assesses students' ability to convert between different forms of rational numbers (fractions, decimals, and percentages). Have students work individually as a formative assessment on the learning targets.



2.3.5 Wrap-Up: Ticket Out the Door

1. Rewrite each value as an equivalent percentage.

- a. 0.02 **2%** b. $\frac{1}{5}$ **20%** c. $\frac{11}{8}$ **137.5%**



Students who write 72.7% as their response for $\frac{11}{8}$ in question 1 inverted the dividend and divisor when calculating the percentage.

2. Rewrite each value as an equivalent fraction.

- a. 67% $\frac{67}{100}$ b. 0.375 $\frac{3}{8}$ c. 35.65 **35** $\frac{13}{20}$

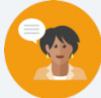


Notice the use of the thousandths place in 0.375 for question 2. Identify students who write $\frac{37.5}{100}$ as well as students who write $\frac{375}{1000}$. Have those students compare their responses to determine why they are equivalent, or highlight these responses in the Warm-Up for Lesson 4.

2.4 Using Operations to Solve Real-World Problems – Part 1

Lesson Introduction

 Benchmark	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		 Learning Target	I can determine which operation should be used to solve a real-world problem.
 Benchmark Coherence	Building On MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.		Working Toward MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- What information do you need to determine the mathematical operation required to create a numeric expression? - How could you use a mathematical operation to represent a real-world scenario?			
 Lesson Narrative	In this lesson, students develop their ability to interpret word problems and determine the operations needed to solve these problems. Using real-world problems with positive decimals and fractions, including mixed numbers, students explore how to sketch or explain their reasoning and create numeric expressions. Students then use developed problem-solving strategies to identify which operations to use to solve real-world problems in the Guided Practice. The Wrap-Up provides students with an opportunity to reflect on learning and to rate their confidence in interpreting which operation should be used for a real-world problem.			
 Component Summary	2.4.1 Warm-Up (~5 min)	Error analysis on discount prices (<i>SE p. 56</i>)	2.4.3 Guided Practice (20 – 25 min)	Practice identifying the appropriate operations required to solve real-world problems (<i>SE p. 58</i>)
	2.4.2 Guided Instruction (20 – 25 min)	Real-world scenarios to determine the appropriate operations for which to write expressions (<i>SE p. 56</i>)	2.4.4 Wrap-Up (~5 min)	Students determine appropriate operation and rate confidence in their answers (<i>SE p. 60</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Numeric expression - Operation		(Optional) Chart paper (Optional) Highlighters (Optional) Markers	
Additional Preparation (Optional) If creating the class anchor chart for 2.4.2 Guided Instruction, label chart paper with one operation each at the top prior to the lesson. Gather markers for students to fill in the chart paper with common key words/phrases used to determine the required operation.				

 Teacher Tips	Common Misconceptions - Students may use the incorrect operation when translating real-world scenarios into numeric expressions. - Students may create inaccurate numeric expressions. - Students may write square inches for the units when evaluating the problems involving inches.	Things to Consider - Position students to recall numeric expressions by having them draw a visual model first and then create their numeric expressions from the visual model. - Students should have experience performing operations on problems involving multi-digit decimals and fractions. Consider opportunities for brief algorithm reviews during the lesson if needed. - If time is an issue, consider assigning some questions from 2.4.3 Guided Practice for homework.
	Literacy and Language Routines Contemplate, Then Calculate Lessons 4-7 in this unit emphasize real-world contexts. Utilize Lesson 4 as a training ground for the process rather than the product of solving, highlighting, and recognizing students' thinking processes rather than their answers. 2.4.3 Guided Practice provides a great opportunity for this using the Contemplate, Then Calculate routine.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts Consider highlighting clarification language in MA.K12.MTR.7.1 that can support the analysis and strategies for problem-solving. - Connect...to everyday experiences - Redesign models or methods to improve accuracy... ...validate conclusions by comparing... ...question the accuracy of models / methods...
	 Differentiate Instruction	

Lesson Notes:

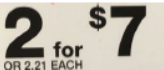
2.4.1 Warm-Up: Math Flop

Component Overview: Consider having students Think-Pair-Share as they analyze the sales tags and notice errors. As students share to the whole group, students can show whether they agree or disagree to engage in discussion as the errors are identified and fixed.



2.4.1 Warm-Up: Math Flop

1. Determine the mistake in each picture. Then, complete the table.

Math Flop	The Mistake: Which number is not correct?	My Work: Show the math that proves it is a mistake.	The Correct Label: Fix the label by marking out the mistake and inserting the correct number.
	Save \$1.49	$\begin{array}{r} 3.99 \\ -2.99 \\ \hline 1.00 \end{array}$	
	Or \$2.21 each	$\frac{7}{2} = 3.50$	



Think-Pair-Share

Consider having students first individually think about and work on the columns in the table as they analyze the errors, then take 1 to 2 minutes to share their responses or progress with a partner. As partners share, listen for strategies or ideas that accurately identify the mistakes, then select these partners to share their thoughts with the class.

2.4.2 Guided Instruction: Interpreting Real-World Problems

Component Overview: Work as a class through questions 1a-1e by drawing the pictures and writing the operations together. Discussions should focus on how students know which operation to use. The first section focuses on identifying and justifying the required operation only, then the second section progresses to creating numeric expressions given real-world expressions. Have students work individually or in partners to write the numerical expressions as they complete the table in question 1f, while circulating the room to check work for accuracy and answering questions.



2.4.2 Guided Instruction: Interpreting Real-World Problems

1. Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in.

Read each question and complete the following table.

- a. Which mathematical operation can be used to find the total length of the two boards?

Operation	Sketch a Picture or Explain Why You Chose that Operation.
Addition	Answers will vary. The word total is a key word that means to add. When you are finding a total, you combine the two amounts, and to combine two totals, you add.

- b. Which mathematical operation can be used to find the length of each piece when the longest board is cut into four equal-sized pieces?

Operation	Sketch a Picture or Explain Why You Chose that Operation.
Division	Answers will vary. The word each is a key word that means division. The problem also states that the board needs to be in 4 equal-sized pieces which indicates the board should be divided by 4.

- c. Check your answers with your partner and resolve any differences if necessary.
- d. Which mathematical operation can be used to determine the difference in the board lengths?
Subtraction
- e. Which mathematical operation can be used to find the total length of 48 of the shortest boards lined up end-to-end?
Multiplication



Avoid teaching key words. This is an unreliable trick that should be avoided. Instead, have students use the strategies built in the lessons to think through the process. Sketching pictures or verbalizing a context would be helpful.



Does your classroom utilize a specific problem-solving strategy? How do your students incorporate MA.K12.MTR.7.1 strategies in practice for real-world contexts? Consider modeling a Think Aloud for question 1a using prompts that reflect your classroom strategies.

- What is the question asking me to do?
- Can I make a sketch of the context?
- Can I restate the problem and my approach to solving it in my own words?
- Can I explain the context to a 5th grader?
- Does this answer seem reasonable?

2.4.2 Guided Instruction: Interpreting Real-World Problems (continued)

- f. Write the numeric expression that shows how to calculate the answer. Then, determine the units of the answer.

Information	Numeric expression	Units of the answer
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the total length of the two boards?	$27\frac{3}{8} + 14\frac{5}{12}$	<i>inches</i>
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the length of each piece when the longest board is cut into four equal-sized pieces?	$27\frac{3}{8} \div 4$	<i>inches</i>
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the difference between the board lengths?	$27\frac{3}{8} - 14\frac{5}{12}$	<i>inches</i>
Louis has two boards. The length of the longer board is $27\frac{3}{8}$ inches (in). The length of the shorter board is $14\frac{5}{12}$ in. What is the total length of 48 of the shortest boards lined up end-to-end?	$48 \cdot 14\frac{5}{12}$	<i>inches</i>



Operations with fractions may be a barrier to learning in this activity.

Consider having students cover up the value to draw a picture then label the picture. Another possibility is to replace fractions with whole numbers for a lower entry point to remove barriers from certain groups of students and then re-establishing the context with fractions.



Notice that students are not actually calculating the solution in order to center the focus of the teaching and learning on the interpretation and approach to solving.



If students are working in small groups or on their own for the final portion of the table, consider challenging some students to actually find the solution for Louis.

2.4.3 Guided Practice: Operations in Real-World Context

Component Overview: Students identify which question uses the correct operation to describe a real-world context.



2.4.3 Guided Practice: Operations in Real-World Context

- The table contains four different real-world contexts. Select the question in the middle column that requires the use of the operation indicated in the last column within this context.

Real-World Context	Which Question?	Operation
A Valencia orange tree that is twelve years old can produce 1.4 boxes of oranges. In May 2017, a box of fresh Valencia oranges cost \$20.45.	☐ For how many years did a box of Valencia oranges cost \$20.45? ☐ How much money would a 12-year old Valencia orange tree generate? ☐ What is the price per box of Valencia oranges?	$\times$
In 2016, there were 19.3 million Snapchat users in the age range of 18-24, and 16.7 million users in the age range of 25-34.	☐ How many Snapchat users were there between the age of 18 and 34? ☐ What was the difference between the number of Snapchat users in the 18-24 and the 25-34 age range? ☐ How many Snapchat users were younger than 18?	$+$



Contemplate, Then Calculate

Constructing an approach or strategy before solving may not come naturally to many students who are accustomed to “just give me the answer” math. Consider approaching this activity with the “Contemplate, Then Calculate” language routine that encourages students to think before doing. Instead of having students actually calculate, the student should state what operation is needed to calculate.



Think-Aloud Sample: Snapchat

- Which question would require me to add something?
- What is the information in the question?
- If there are 19.3 million in one range and 16.7 million in the other range, maybe ask a question about both ranges together or total.
- “How many users between 18 and 34?” This is both ranges total, so yes, I would have to add to find both ranges.

2.4.3 Guided Practice: Operations in Real-World Context (continued)

In 2004, each person in the United States consumed 61.5 pounds of refined sugar on average. In 2008, the average consumed per person was 66.3 pounds of refined sugar.	□ What is the total amount of sugar the average person consumed in 2004 and 2008? □ What was the average yearly increase in sugar consumption from 2004 to 2008? □ What is the difference between the amount of sugar consumed in 2004 and 2008?	—
Barbara sent letters that exceeded the allowed weight by the U.S. Postal Service. To mail the letters she had to add a \$0.15 stamp to each letter. She spent \$2.55 in additional weight charges.	□ What was the total cost of sending the letters? □ For how many letters did Barbara pay the additional weight charge? □ What is the difference between the cost of sending a regular letter and one that exceeds the weight allowed?	÷



If utilizing small-group learning, consider having students take turns modeling a Think Aloud for each other, illuminating different ways of thinking and entry points for different types of learners.

2.4.3 Guided Practice: Operations in Real-World Context (continued)

2. Use the same four real-world contexts and the operation indicated to find the answer to the question selected in the previous table. Write the numeric expression used to determine the answer. Then, show your work and write the answer using the correct units.

Information	Numeric expression	Show your work and answer with units
A Valencia orange tree that is 12 years old can produce 1.4 boxes of oranges. In May 2017, a box of fresh Valencia oranges cost \$20.45.	$20.45 \cdot 1.4$	\$28.63
In 2016, there were 19.3 million Snapchat users in the age range of 18-24 and 16.7 million users in the age range of 25-34.	$19.3 + 16.7$	36 million
In 2004, each person in the United States consumed 61.5 pounds of refined sugar on average. In 2008, the average consumed per person was 66.3 pounds of refined sugar.	$66.3 - 61.5$	4.8 pounds
Barbara sent letters that exceeded the allowed weight by the U.S. Postal Service. To mail the letters, she had to add a \$0.15 stamp to each letter. She spent \$2.55 in additional weight charges.	$2.55 \div 0.15$	17 letters



Consider putting students into groups of four. Each student has their own problem to solve and should share their thinking with the other students in their group. The teacher then can call on different groups for the person that had the problem to share their thinking with the whole class.



To extend this activity, students who finish early can create their own real-world contexts, numeric expressions, and answers with correct units to add to the table. Consider asking these students to share their work with a partner to check for accuracy.

2.4.4 Wrap-Up: Reflection

Component Overview: As an assessment of learning targets, students individually determine the operation needed for given real-world scenarios, then rate their confidence in the accuracy of their answer choices. Data and observations from the lesson and the Wrap-Up can be used to determine where students stand for the next lessons in this unit. Students should not use a calculator to complete the Wrap-Up.



2.4.4 Wrap-Up: Reflection

Mathematics is most useful in the real world. Determine the operation(s) needed to find the answer to each question. Then, mark how confident you are in your choice of operation(s) to be used.

1. Fitzgerald is selling chocolate bars to raise money for a trip. He has to sell a total of 4 boxes. The first day, he sold $2\frac{3}{4}$ boxes. How many more boxes does he have to sell?

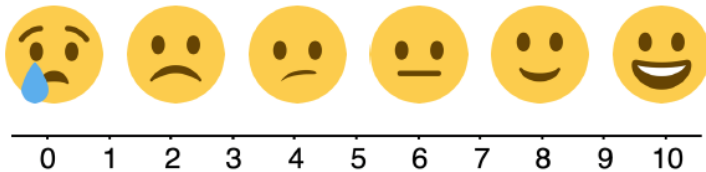
The operation(s) that should be used is/are
subtraction.



Answers will vary.

2. Ginger is buying new carpet for her living room. The carpet costs \$56.81 per square yard with an additional \$7.64 per square yard to install it. The size of the living room is 13.6 square yards. How much will the carpet for the living room cost?

The operation(s) that should be used is/are
multiplication and addition.



Answers will vary.



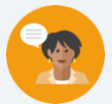
Considering the variety of correct or incorrect responses for each prompt and what those responses are, along with the emoji scale, can tell you about learning progression.

Question 1: Students could choose addition if they estimate $1\frac{1}{4}$ to then add to $2\frac{3}{4}$. Both addition or subtraction could be correct and will tell you how that student is approaching problem-solving.

2.5 Using Operations to Solve Real-World Problems – Part 2

Lesson Introduction

Lesson Introduction					
 Benchmark	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		 Learning Target	I can solve real-world problems with up to two steps using decimals or fractions.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.		MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.		
 Essential Questions	- What financial terms are associated with the four operations? - How do you know which operation to use in real-world problems?				
 Lesson Narrative	In this lesson, students solve one-step problems involving money with decimals in a real-world finance context. In the guided sections, students learn about financial terms and complete activities that form deeper problem-solving connections and strengthen operational fluency using real-world situations. Students then use this practice to complete one-step problems involving real-world finance situations.				
 Component Summary	2.5.1 Warm-Up (~10 min)	Budgeting video and discussion (<i>SE p. 61</i>)	2.5.3 Guided Practice (15-20 min)	Practice completing one-step problems involving real-world finance situations (<i>SE p. 63</i>)	
	2.5.2 Guided Instruction (20-25 min)	Guided introduction to financial terms and real-world problems (<i>SE p. 61</i>)	2.5.4 Wrap-Up (5-10 min)	Move on to solving one-step problems involving real-world finance situations (<i>SE p. 64</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Income - Finance - Investment - Principal - Return - Gross income - Net income Additional Terms: N/A		(Optional) Chart paper (Optional) Markers		2.5.1 Warm-Up contains a video to accompany the question prompts. (Optional): Create table for problems in 2.5.3 Guided Practice. (Optional): Print or prepare students to create Frayer Models or an organized table for the vocabulary words in this lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may confuse financial terms with their mathematical impact or meaning, like investment and return.	- Consider “pre-work” for this lesson as the previous day’s homework to better prepare students to navigate financial terms for this lesson (students can watch a video explaining these terms as “pre-work” or completing question 3 in 2.5.2 as “pre-work” for the lesson using an online dictionary). - Consider providing laminated sample budgets, checks, and paychecks for students to reference during the lesson. Circle values that align with the terms.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads 2.5.2 Guided Instruction provides numerous opportunities for students to utilize reading comprehension skills in a real-world mathematic scenario. Consider applying “Three Reads” routines throughout the activity, with an emphasis on problems with higher complexity. Students can also write new financial terms, definitions, examples, and non-examples in journals following the “Three Reads”.	MA.K12.MTR.5.1: Patterns and Structure Students will be introduced to financial terms that may be unfamiliar. Encourage students to look for similarities and patterns when solving real-world finance problems that focus on relevant words/phrases, creating a plan to solve, and relate financial terms to concepts in mathematical operations. Students can connect strategies learned in Lesson 4 to the finance-specific problems of this lesson.
 Differentiate Instruction	Students are introduced to potentially new financial terms. Consider having students create a graphic organizer, like a table of Frayer Models for each vocabulary words, from 2.5.3 Guided Practice. The table can be drawn in math journals or on chart paper with the word, operation, examples, and non-examples, as a visual aid for problem scaffolding.	
	Consider a variety of methods for students to demonstrate or express their learning, using projects like budget sheets, balancing the payroll, small business CEO, and more to engage students in multiple representations of their learning.	
Lesson Notes:		

2.5.1 Warm-Up: Two Questions and One Wonder

Component Overview: Display the Warm-Up 4-minute video for the entire class. In the video, Ky-ree, a firefighter in Los Angeles, discusses budgeting his salary and monthly paycheck.



2.5.1 Warm-Up: Two Questions and One Wonder

Ky-ree, a firefighter in Los Angeles, budgets his paycheck.

Ky-ree's Budget	
Monthly salary	\$5,800
Taxes and other deductions	\$2,900
Net take home income	\$2,900
Rent	\$1,350
Utilities	\$60
Internet & cable	\$50
Cell phone	\$130
Car payment	\$330
Insurance	\$120
Gas	\$140
Food	\$250
Gym	\$30
Entertainment	\$120
Remaining	\$320

- List two questions you have about Ky-ree's budget.
Answers will vary. What do utilities include? What about water and electric?
- List one thing you wonder about Ky-ree's budget.
Answers will vary. I wonder what is being added or subtracted.



Utilize this discussion as an opportunity to begin incorporating the math vocabulary of this lesson in context, even referencing or beginning to fill in the Word Bank in 2.5.2 Guided Instruction to orient students to the content.



As an extension for project-based learning, consider asking other adults with various types of jobs (or have students research other salaries) to share an example of their monthly budget. These budgets can be laminated and used by students to help them get familiar with the vocabulary.

2.5.2 Guided Instruction: Real-World Contexts About Finances

Component Overview: Students are introduced to financial terms and complete activities designed to provide opportunities to form deeper connections using real-world situations.



2.5.2 Guided Instruction: Real-World Contexts About Finances

Income is the amount of money received.

- Recall the problem from Lesson 4.
A Valencia orange tree that is 12 years old can produce 1.4 boxes of fruit. In May 2017, a box of fresh Valencia oranges cost \$20.45. How much income would a Valencia orange tree generate?
\$28.63 of income

Finance is the management of money.

- Daniel owns a Valencia orange grove. Choose other items from the list that he should consider as a part of his finances.
 - ☐ The taxes he pays on the property.
 - ☐ The amount of money he has in the bank.
 - ☐ The cost of taking care of the orange grove.
- Use the words in the word bank to complete each statement.

Word Bank

withdrawal	checking	costs	deposit
expenses	savings	taxes	

- Costs** or **expenses** describe money that is spent.
- Taxes** are the amount of money that is paid to the government.
- A **deposit** is the amount of money placed in a bank account.
- A **withdrawal** is the amount of money removed from a bank account.



Three Reads

Consider modeling the Three Reads routine for students to later apply in Guided Practice, targeting the interpretation and translation of verbal to algebraic descriptions.

- First Read:** Reading through the prompt to get familiar with the context.
- Second Read:** Reading through the prompt specifically looking for important information to circle, underline, or note.
- Third Read:** Read the question and look for information within the prompt to answer the question.



As financial terms, definitions, and examples are introduced, consider having students read and write the definitions in a math journal using a table and/or Frayer Model.



Consider placing students in groups of 2 to 3 to complete question 3 and discuss options. Encourage small groups to discuss how and why they chose the words they did to fill in the blanks. Ensure that the students have chosen the correct terms, as a firm understanding of these finance terms is important to problem solving in upcoming activities.

2.5.2 Guided Instruction: Real-World Contexts About Finances (continued)

- e. **Checking** and **savings** are types of bank accounts.

An **investment** is when the initial amount of money, known as the **principal**, earns more money, known as the **return**, over time.

4. For each financial transaction, write the mathematical operation that is most commonly associated with it.

Financial Transaction	Mathematical Operation
costs	—
deposit	+
expenses	—
income	+
taxes	—
withdrawal	—
return on an investment	+



Consider reproducing this table and adding other key terms as an anchor chart for interpreting mathematical phrases in real-world contexts for students to use throughout this unit. Encourage a variety of depictions, including verbal descriptions, drawn pictures, printed images, and more.

Gross income is the amount of income that is earned.

5. Hannah earns \$9.17 an hour. Each week she works 14 hours. How much is Hannah's weekly gross income?
\$128.38 per week

Net income is the amount of income that is earned after expenses and taxes are deducted.

6. Each week Hannah has to pay \$26.96 in taxes. What is Hannah's net income?
\$101.42 per week

2.5.3 Guided Practice: Solving Real-World Finance Problems

Component Overview: Students solve one-step problems involving real-world finance situations. Consider working as a class for question 1 to model a “Think Aloud” and how to use visual diagrams to support problem solving, then having students work independently or in small groups for questions 2-4. Remind students to use the table and definitions in previous sections to work out the problems.



2.5.3 Guided Practice: Solving Real-World Finance Problems

1. A trampoline park charges \$26.88 for 2 hours.
 - a. How much is the cost for 1 hour?
\$13.44 per hour
 - b. The manager of the trampoline park knows that during a Saturday there are typically 125 people at the park. If each person only pays for 2 hours, what is the park’s typical income on Saturdays?
\$3,360.00
 - c. The manager of the trampoline park lists the costs for Saturday as follows: \$119.50 to pay the employees and \$256.73 for all other costs. What are the park’s typical costs?
\$376.23
 - d. Determine the park’s net income for a typical Saturday.
\$2,983.77
2. Emmanuel invested \$12,500 into the stock market. His return, the money he earned on his investment, was \$281.25 for each of the first three quarters of the year. He combines his returns with his initial investment. How much does Emmanuel have invested after the third quarter of the year?
\$13,343.75
3. Aaron has \$2,478.23 in his checking account. He writes a check for \$637.89 to pay for new tires on his truck. How much money does he have in his checking account now?
\$1,840.34
4. Violeta has \$5,734.49 in her savings account. She deposits \$218.56 and withdraws \$92.50. How much does she have in her savings account now?
\$5,860.55



Consider providing a pre-constructed table for scaffolding practice or encouraging students to model their own strategies for making sense of the concepts when practicing MA.K12.MTR.7.1.

Example:

What is the question asking?	Operation ($\pm \times \div$)	My Approach / Strategy



For a visual model, in question 1d, draw a bar diagram that shows the total costs in sections, including the unknown of the net income that is determined through subtraction.



Three Reads

As students work in groups to work on questions 2-4, have students use Three Reads to make sure they understand what they are trying to find in relation to financial terms. Students can highlight, circle, or underline vocabulary words that they notice.



Challenge students to create new financial problems that use the information given in the existing problems or from Ky-ree’s budget in the Warm-Up. The new problems can be one step and real-world, then have students work in small groups to solve the problems.

2.5.4 Wrap-Up: Ticket Out the Door

Component Overview: Students work individually to use a given budget for a music festival to answer a series of questions and solve a problem related to the planning. Student understanding of financial terms learned in this lesson is also assessed through embedded questioning. Students should not use a calculator to complete the Wrap-Up.



2.5.4 Wrap-Up: Ticket Out the Door

Mishka is planning to host a music festival to raise funds for a local food bank. Her budget is shown.

Item	Amount
Ticket price at the door	\$9.00
Fee paid to each band	\$175.00
Early-bird ticket price	\$6.50
Equipment rental	\$225.00
Publicity (flyers/artwork)	\$64.57
Stage rental	\$187.50

- In order to find the **gross income** for the music festival, I need to know the number of people who bought an early bird ticket and the number of people who bought a ticket at the gate.
- List the items in Mishka's budget that are **expenses**.
Band fee, equipment rental, publicity, and stage rental
- If Mishka had a gross income of \$1,166.50 and expenses totaling \$827.07, what is her net income for the music festival?
\$339.43



Challenge students by incorporating new financial terms like revenue and profit to re-word problems from the activities in the lesson.

Example: After defining revenue and profit, ask, "How are gross income and net income related to revenue and profit?" or "Write your own word problem about the Valencia Grove using key terms from the lesson."

2.6 Solving Two-Step, Real-World Problems With Fractions or Decimals

Lesson Introduction

 Benchmark	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		 Learning Target	I can solve real-world problems with two steps using decimals or fractions.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.		MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- What are the similarities and differences between solving one-step and two-step real-world problems? - What different strategies can be used to solve two-step equations with decimals and fractions?			
 Lesson Narrative	In this lesson, students continue applying their knowledge of operations on rational numbers to solve two-step problems. Then, students collaborate by using feedback to improve on their solutions. Students then use the problem-solving process and skills from previous lessons, including converting rational numbers and writing numeric expressions for real-world problems to solve two-step real-world problems involving decimals and fractions.			
 Component Summary	2.6.1 Warm-Up (~5 min.)	Would You Rather Activity (SE p. 65)	2.6.3 Collaboration (10-15 min.)	Error analysis, correction, and growth mindset message on two-step, real-world problems (SE p. 66)
	2.6.2 Guided Practice (10-15 min)	Practice solving two-step equations through scaffolded problems (SE p. 65)	2.6.4 Your Turn! (15-20 min.)	Practice solving two-step, real-world problems (optional Gallery Walk opportunity) (SE p. 67)
	2.6.5 Wrap-Up (~5 min)	Confidence scale for solving real-world problems with decimals and fractions (SE p. 68)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Markers	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions • Students may create inaccurate numeric expressions. • Students may make computational errors when performing operations on multi-digit decimal and fraction problems. • Students may incorrectly round their answer for money problems.	Things to Consider • To foster flexible thinking and problem-solving strategies, ask students to draw a visual model first and then to create a numeric expression from the visual model. • Students may need to round to the nearest hundredth for money problems so that solutions make sense. • Students have additional practice for fluency with decimals and fractions throughout the rest of the unit. Leverage the 2.6.5 Wrap-Up in this lesson as an opportunity to facilitate growth mindset and self-reflection. • Encourage students to use problem-solving strategies and not processes that involve using only keywords or shortcuts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk 2.6.1 Warm-Up offers a unique opportunity for students to engage in meaningful mathematical discourse in a friendly way – with Skittles. Consider what this entry point (even with real Skittles) could do for activating students’ engagement with decimals prior to the lesson.	MA.K12.MTR.4.1: Discussion and Reflection In 2.6.3 Collaboration, students analyze the mathematical thinking of another student. Consider the clarification language in MA.K12.MTR.4.1 and how that can help frame important mathematical dialogue as students discuss their thinking, sequence and present their findings, and develop methods of comparison and justification when analyzing work.
 Differentiate Instruction	For 2.6.4 Your Turn!, questions 1 and 3 use fractions while questions 2 and 4 demonstrate decimal fluency. Consider how to best engage your students in sequencing which questions they work on and in what order. Students could complete one of each before comparing with a peer or students could engage in choosing which they feel least or most confident in first.	
	For 2.6.4 Your Turn!, encourage visual learners to draw models prior to creating numerical expressions. For students who are struggling with the computations involving fraction or decimal parts, consider replacing the fractions and decimals with whole numbers, allowing them to focus on understanding the problem-solving process rather than the computation.	
Lesson Notes:		

2.6.1 Warm-Up: Would You Rather?

Component Overview: Students compare two different deals for Skittles to determine the best purchase. Consider using Think Pair Share literacy routines to give students to work individually then hear others' strategies or ideas in a class discussion.



2.6.1 Warm-Up: Would You Rather?

Would you rather buy 8 Skittles for 25 cents or 62 Skittles for \$1.49?



1. Explain your choice.
Answers will vary.
2. Would your answer change if you could choose what flavor the 8 Skittles were but you could not choose the flavors from the bag?
Answers will vary.



Students should be applying skills of estimation in ranges of fractional or decimal quantities. Consider leveraging some of their responses from this activity later in the lesson when grappling with similar fraction and decimal quantities.

- “I figured there are 8 times as many skittles in the second offer, but the price is less than 8 times the 25 cents.” (ratios)
- “I used division to figure out how much 1 skittle cost in each, with the second offer being less cost per skittle.” (unit rate)

2.6.2 Guided Practice: Two-Step Rational Problems

Component Overview: Students practice solving two-step equations through scaffolded problems. Consider having students work in small groups or partnerships to engage in math discussions and check work for accuracy for question 1, then work individually to complete question 2. After question 1, engage in class discussion on strategies used and compare strategies for effectiveness.



2.6.2 Guided Practice: Two-Step Rational Problems

1. Sophie grew her hair out for a long period of time. Her hair is $2\frac{3}{5}$ feet long. She donated her hair by cutting $\frac{5}{6}$ of a foot off. Sophie's hair then grew $1\frac{2}{3}$ feet.
 - a. What operations are needed to determine the length of Sophie's hair?
Addition and Subtraction
 - b. What values are necessary to determine the length of Sophie's hair?
 $2\frac{3}{5}, \frac{5}{6}, 1\frac{2}{3}$
 - c. Write the numeric expression that can be used to determine the length of Sophie's hair.
 $2\frac{3}{5} - \frac{5}{6} + 1\frac{2}{3}$
 - d. Determine the length of Sophie's hair.
 $3\frac{13}{30}$ ft
 - e. Determine if the length of Sophie's hair makes sense. Support your answer.
Answers will vary. No, the length of Sophie's hair does not make sense. It seems too long.
2. Tiffany has a balance of \$167.32 in her checking account. She deposits \$58.16 in her checking account and then writes a check for \$97.27.
 - a. What operations are needed to determine the amount of money in Tiffany's bank account?
Addition and Subtraction
 - b. What values are necessary to determine the amount of money in Tiffany's bank account?
 $\$167.32, \$58.16, \$97.27$
 - c. Write the numeric expression that can be used to determine the amount of money in Tiffany's bank account.
 $\$167.32 + \$58.16 - \$97.27$
 - d. Determine the amount of money Tiffany has left in her checking account.
 $\$128.21$



Notice the inherently scaffolded prompts in this Guided Practice. While monitoring students' practice, consider asking probing questions as informal observational data for the remainder of the lesson for the following questions.

- Question 1a: How do you know? What will that look like for letter C?
- Question 1b: What do you notice about the fractions? What does that make you think about when adding and subtracting those values?
- Question 1d: Does that mean your answer is wrong? Why or why not?



Consider providing grid or graph paper for students to organize their work when calculating with decimals.



Though these problems involve fluency with decimal and fraction operations, position students to focus on the lesson's learning targets as they complete these problems individually. To facilitate this focus, ask questions like:

- How do you know what operation is required?
- How do you know how many steps are involved in finding the final answer?
- How can you check your answer for feasibility within the given context?

2.6.3 Collaboration: Error Analysis

Component Overview: Students complete an Error Analysis task where they analyze and explain the errors, then provide a brief note explaining the mistake and how to learn from it. Group students in partnerships as they evaluate the work, then complete the problem correctly and provide reasons for their solution. Consider combining partnerships for the feedback in question 1c so students share their thinking with others.



2.6.3 Collaboration: Error Analysis

1. Stefanie solved the following word problem. Her work is shown.

- a. Evaluate her work and determine where she made mistakes.

Problem	Stefanie's Work
Janelle has $2\frac{1}{4}$ pans of brownies. Each pan made 8 brownies. She gave $\frac{2}{3}$ of the brownies to her friends. How many brownies did Janelle give away?	Step 1: $2\frac{1}{4} \times 8 = \frac{9}{4} \times 8 = \frac{18}{4}$ Step 2: $\frac{18}{4} \div \frac{2}{3} = \frac{18}{4} \times \frac{3}{2} = 6\frac{3}{4}$ Answer: She gave away $6\frac{3}{4}$ brownies.

- b. Help Stefanie understand and learn from her mistakes. Write a brief note to Stefanie explaining what her mistakes were and how she could learn from those mistakes.

Answers may vary.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Possible answer:

When finding a fraction of an amount, use multiplication. When multiplying a fraction by a whole number, change the whole number to a fraction with denominator of 1.

Solution: $(2\frac{1}{4} \times 8) \times \frac{2}{3} = 12$

- c. Read your partner's note to Stefanie and offer suggestions for how to improve their note.



MA.K12.MTR.4.1: Discussion and Reflection
Stefanie makes a computational error in Step 1 when determining how many total brownies are available to give away. Encourage students not only to identify the error, but also to highlight the correct use of multiplication to determine the number of brownies, as well as the opportunity to assess that $\frac{18}{4}$ brownies was not reasonable given the context.



Consider the following probing questions as students collaborate on identifying the error:

- Why is $2\frac{1}{4}$ multiplied by 8? Does the answer $\frac{18}{4}$ make sense? Why or why not?
- Why is that product then divided by $\frac{2}{3}$? Why not addition or multiplication?
- Does the answer seem reasonable?

2.6.4 Your Turn!

Component Overview: Consider using a Gallery Walk for this activity. Provide small groups of 2 to 4 with chart paper and markers. Have students solve all four problems on their chart paper. After about 15 minutes of small group work time, have students display their chart paper on a visible wall/desk, then have the class walk around to view other groups' posters for about 5 minutes. Ask students to visit at least three posters and confirm the work on the poster with their own answer and compare strategies used.



2.6.4 Your Turn!

1. Martha fills her ice cream bowl with $\frac{3}{4}$ cup of ice cream, $\frac{1}{8}$ cup of candy toppings, and the rest with fruit toppings. If the bowl contains a total of $1\frac{1}{2}$ cups, how much of the bowl is made up of fruit toppings?

$\frac{5}{8}$ cups of fruit topping

2. Jasmine has a ribbon that is 12.6 meters long. Sharon has a ribbon that is 3.8 meters shorter. What is the total length of the two ribbons?

21.4 meters

3. Before soccer practice, Hernando puts $4\frac{1}{8}$ cups of sports drink in a water bottle with $1\frac{2}{3}$ cups of ice. After a hot afternoon of practice, he drinks $2\frac{1}{5}$ cups of the sports drink and melted ice. How many cups of this mixture are left in the water bottle?

$3\frac{71}{120}$ cups

4. A bag of chocolates weighs 1.384 pounds. If Luis gives each of his 6 friends 0.23 pounds of chocolate, how many pounds of chocolate does Luis have left?

0.004 pounds



Encourage students to draw visual models prior to attempting to solve the problem. This may help students make connections from a pictorial representation to the numeric expression and operation needed to solve.



To encourage mathematical reasoning, ask questions like "Do your answers make sense? What methods can you use to check your answer?" (MA.K12.MTR.7.1)



As students work on answering the questions, circulate around the room and listen for how students are making sense of them. To develop strategies for understanding problems prior to solving, ask questions like:

- How do you know what you are being asked to find?
- Which step/operation do you need to use first? Which operation/step do you use second? How do you know?
- What strategy can you use? Does drawing a model make sense for this problem?
- Is your answer reasonable? How do you know?

2.6.5 Wrap-Up: Reflection

Component Overview: Students rate confidence with solving real-world, two-step word problems. Student responses and class observations collected from the lesson combined with this reflection to assess student understanding for small group intervention.



2.6.5 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can solve real-world problems with two steps using decimals.

Answers will vary.

I need help. I can't get started. I'm getting there. I understand and have evidence.



2. I can solve real-world problems with two steps using fractions.

Answers will vary.

I need help. I can't get started. I'm getting there. I understand and have evidence.



Consider scaffolded prompts for students to more accurately reflect on their progress.

- *I need help: "I don't know where to start" or "Decimals and fractions are difficult for me"*
- *I'm getting there: "I get some problems right and some wrong" or "I know how to solve but sometimes struggle with decimals"*
- *I understand and have evidence: "I could explain how to do this to someone else" or "Provide an example to show evidence"*

2.7 Solving Multi-Step, Real-World Problems With Fractions or Decimals

Lesson Introduction

 Benchmark	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		 Learning Target	I can solve real-world problems with three steps using decimals or fractions.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.		MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- What are the similarities and differences between solving two-step and three-step real-world problems? - How can you tell what steps and operations are needed to solve three-step problems? - How do you know if your answer to a multi-step problem is reasonable?			
 Lesson Narrative	In this lesson, students continue to solve real-world, multi-step word problems involving rational numbers, using the problem-solving skills they developed in earlier lessons from this unit. Students are guided through problem solving with three-step real-world problems involving decimals and fractions. In the Collaboration, students specifically apply these skills to analyze the distance between hurdles in a 110-meter hurdle race, then determine which arithmetic operations are relevant and use them to solve the problems. Additionally, students draw or use a diagram to help them make sense of measurements, as well as to communicate their reasoning about the measurements.			
 Component Summary	2.7.1 Warm-Up (~5 min.)	Explore short division method (<i>SE p. 69</i>)	2.7.3 Collaboration (10 min.)	Analyze and solve problems related to a 110-meter hurdle race (<i>SE p. 71</i>)
	2.7.2 Guided Instruction (15-20 min.)	Guided instruction on problem solving with three-steps (<i>SE p. 69</i>)	2.7.4 Your Turn! (5-10 min.)	Practice solving three-step real-world problems with rational numbers (<i>SE p. 72</i>)
	2.7.5 Wrap-Up (~5 min.)	Students rate whether they agree, disagree, or are unsure about statements and justify their thinking (<i>SE p. 72</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Markers	
		Additional Preparation		
		If you plan to go to the school track, prepare hurdles or something that will mimic the hurdles and measuring tools.		

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may create inaccurate numeric expressions. - Students may make computational errors when performing operations on multi-digit decimal and fraction problems. - Students may have difficulty identifying the steps needed to solve problems, and in which order to solve the steps to find the correct solution. - In 2.7.3 Collaboration, students may create 10 spaces (instead of 9) between 10 hurdles, leading to a miscalculation of the distance between hurdles and incorrect division in the following steps.	- To foster flexible thinking and problem-solving strategies, ask students to draw a visual model first and then create a numeric expression from the visual model. - Provide students who may need assistance organizing information a graphic organizer and/or annotation strategies while engaging with three-step problems. - For the context of the 110-meter hurdles in 2.7.3 Collaboration, the distance between the first and last hurdles is 82.26 meters. If students end up with a non-terminating decimal, have them to revisit each step to find the error. - If time is an issue, then consider assigning 2.7.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Utilize Think-Pair-Share opportunities in 2.7.2 Guided Instruction and 2.7.4 Your Turn! for students to practice articulating their strategies and approaches in problem-solving in order to continue building fluency in real-world contexts.	MA.K12.MTR.7.1: Real-World Contexts Throughout the lesson, students will be reading, analyzing, and discussing real-world scenarios to determine the appropriate mathematical operation to apply for solving problems. Students use concrete and abstract models to investigate real-world problems and find solutions that are reasonable given the context.
 Differentiate Instruction	Consider using a jigsaw for 2.7.4 Your Turn! practice problems. Students can be assigned to work on one of the five problems, then rotate to new groups where one student is responsible for sharing thinking and answers for the problem as the other students work on it.	
	Reading comprehension is at the forefront of learning for real-world problems. Consider your students when planning for 2.7.5 Wrap-Up – this might work for your students if they need independent processing time, or this activity could be transformed into a Socratic seminar-style debate with evidence required in collaboration with an English teacher.	
Lesson Notes:		

2.7.1 Warm-Up: Short Division

Component Overview: Students investigate another method of division, short division.



2.7.1 Warm-Up: Short Division

In Mexico, when dividing by a single-digit divisor, sometimes a method called “short division” is used. Consider the example shown of $74 \div 4$.

$$\begin{array}{r} 18 \\ 4 \overline{) 74} \end{array}$$

The quotient 18 is written above the dividend 74. The 1 is blue, the 8 is orange. The 3 in 74 is blue, and the 2 in 4 is orange.

- Using your knowledge of division, explain where you think the 1, 3, 8, and 2 came from in the process.
First, the tens place is divided. 40 divides into 70 only 1 time so that is where the 1 comes from. 3 represents the remainder 30. Next the ones place is divided. 4 goes into 34 8 times and has a remainder of 2.
- Show how you might represent the same division problem using a method different from “short division.”
Answers will vary.



Clarify, Correct, Critique

Students analyze a new method of division, short division. To strengthen metacognitive thinking and problem-solving skills, consider positioning students to clarify, correct, and/or critique their own and others’ reasoning by analyzing the new method, and comparing it to the standard long division algorithm.

2.7.2 Guided Instruction: Three-Step Rational Problems

Component Overview: Students practice planning and solving three-step rational problems. Encourage students to recall strategies used to solve one- and two-step real world problems in previous lessons to build upon background knowledge. Whole group and or small group work can be incorporated in this activity to provide differentiation for different types of learners.



2.7.2 Guided Instruction: Three-Step Rational Problems

- For his recital, a student practiced the piano 2.18 hours each day for 5 days last week. This week, he practiced 0.75 hours each day for 4 days. How many hours did he spend practicing? **13.9 hours**
- The Across Florida 200 Trail Run is an event where participants run 200 miles of trails between the Withlacoochee Bay Trail and Crescent Beach. To prepare for the Across Florida 200 Trail Run, Walt ran 2.18 hours each day for 5 days last week. This week, he ran 2.75 hours each day for 4 days. Walt runs at an average pace of 6.23 miles per hour.
 - How many hours has Walt run over the last two weeks? **21.9 hours**
 - How many miles has Walt run over the last two weeks? **136.437 miles**
 - Based on Walt's average pace, how much longer would it take him to run the Across Florida 200 Trail Run than the amount of time he has spent practicing the last two weeks? **≈ 10.2 hours longer**
- The Across Florida 200 Trail Run offers a team competition where a team of 4 runners take turns to run the 200 miles.

A team of runners and their average pace is shown.

Name	Molly	Reginald	Tyler	Liz
Average Pace (miles per hour)	6.75	6.75	8.2	9.58

To prepare for the race the team ran part of the trail.

Team Member	Description	Miles
Tyler	Withlacoochee Bay Trail to Lake Rousseau Dam & Recreational Area	10
Liz	Lake Rousseau to Florida Trail	15
Molly	Florida Trail Part 1	35
Reginald	Florida Trail Part 2	25

How long did it take the team to run the practice run of 85 miles? **11.7 hours**



Think-Pair-Share

Providing students with ample processing time using Think-Pair-Share strategy gives both independent and verbal processing for more precise and confident understanding.



Real-world problems require reading and comprehension. Consider pairing with a reading teacher for interdisciplinary support. Students can get additional support in identifying key features of the text that will lead to better comprehension, in addition to more practice time. For example, if students use a specific reading comprehension strategy with another teacher, consider having students apply that strategy in these contexts.



Consider prompting questions to guide students toward key features of the text.

- What is the question asking? (Total time ran by all)
- How will we know how long they took? (Add up how long each person took.)
- How do we know how long each person will take? (Multiply their pace times total miles run.)

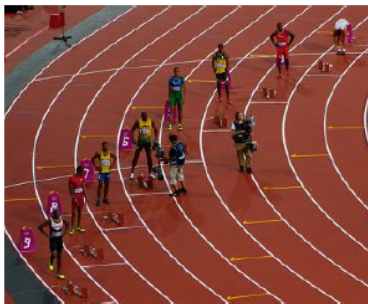
2.7.3 Collaboration: Distance Between Two Hurdles

Component Overview: Students collaborate to analyze and answer questions about the distance between hurdles in a 110-meter hurdle race. Prior to starting the activity, consider showing a video and/or images of 110-meter hurdle races and explain to students that the starting blocks are placed so that the outer and inner lanes still have the same running distance. Consider placing students in small groups based on learning styles and encourage visual and kinesthetic learners to act out or create a model of the scenario. Students then use these models to make sense of the measurements found and communicate their reasoning about the measurements.



2.7.3 Collaboration: Distance Between Two Hurdles

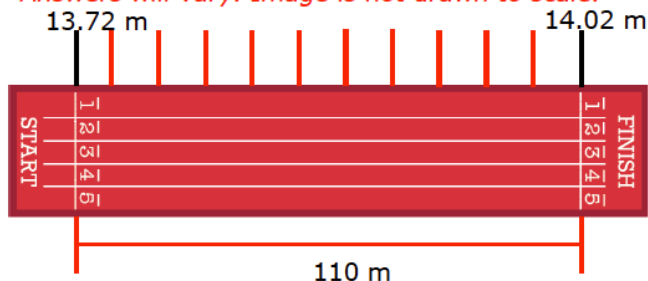
There are 10 equally spaced hurdles on a racetrack. The first hurdle is 13.72 meters from the start line. The final hurdle is 14.02 meters from the finish line. The racetrack is 110 meters long.



The diagram below represents what the track would look like if the track were straight and not curved.

1. Use the diagram below to show the placement of the hurdles on the racetrack. Label all known measurements.

Answers will vary. Image is not drawn to scale.



2. Discuss with your partner how many spaces there are between the 10 hurdles.

9 spaces

3. How far are the hurdles from one another? Explain or show your reasoning.

$$110 - 13.72 - 14.02 = 82.26$$

$$82.26 \div 9 = 9.14$$

9.14 meters



Before students begin, consider discussing how the hurdles are staggered to compensate for the inner lanes being a shorter distance than the outer lanes. Also, discuss why there are 9 spaces between the 10 hurdles.



Do you have track student-athletes in our class, or do you have a track at the school? Consider taking a field trip and implementing this lesson on your school's track outside to activate visual and kinesthetic entry-points.



For questions 2 and 3, if students make arithmetic errors when calculating the distance between the first and last hurdles (82.26 meters), ask students to check their work with questions like:

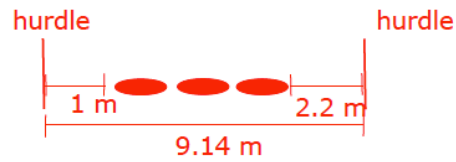
- How many spaces are needed between 2 hurdles? 3 hurdles? 4 hurdles? How can you extend the pattern to 10 hurdles?
- Is your answer reasonable? How can you use estimation and rounding to check?

2.7.3 Collaboration: Distance Between Two Hurdles (continued)

4. A professional runner takes 3 strides between each pair of hurdles. The runner leaves the ground 2.2 meters before the hurdle and returns to the ground 1 meter after the hurdle.

- a. Draw a diagram to model the runner's strides between the hurdles. Label all known measurements.

Answers will vary. Image not drawn to scale.



- b. About how long are each of the runner's strides between the hurdles? Show your reasoning.

$$9.14 - 1 - 2.2 = 5.94$$

$$5.94 \div 3 = 1.98$$

1.98 meters



Clarify, Correct, Critique

As collaborating groups finish and have a completed diagram for question 4a and answer for question 4b, consider combining groups and positioning students to clarify, correct, and/or critique the responses of others. Groups can take turns sharing their diagrams, then the other students can ask clarifying questions, offer suggestions for correction, and critique the diagram for accuracy.

2.7.4 Your Turn!

Component Overview: The unit has used scaffolding to help students become successful at solving problems using rational numbers. Fluency with rational values is an important focus for the 6th grade accelerated course. Teachers may consider allowing students to use a calculator to answer the questions.



2.7.4 Your Turn!

1. A pool that holds 273.68 gallons of water lost 4.79 gallons of water each day while there was a leak that went unnoticed. Tad noticed the water was lower, so he added 20.45 gallons of water before he realized there was a leak on the side. If Tad fixed the leak after 8 days of leaking, how many gallons of water does he currently have in the pool?

255.81 gallons

2. A grocery store opens a new wing bar that charges by the pound. Yasmine wants to purchase $\frac{1}{4}$ pound of garlic parmesan wings, $\frac{1}{8}$ pound of mild wings, and $\frac{2}{5}$ pound of teriyaki wings. She also wants to purchase some hot wings. If Yasmine wants to purchase a total of 2 pounds of wings, how many pounds of hot wings should she buy?

$1\frac{9}{40}$ pounds



For students who finish early, challenge them to determine after how many days the pool would be empty if the leak was not fixed (58 days), and to write an expression that could represent that value.



Did you subtract the amount of each type of wing from 2 pounds? Or did you add up all types before subtracting that value from 2 pounds? Would you do this differently next time? Why or why not?



Differentiate the content to engage more learners by giving students the opportunity to substitute one of the questions with their own question and self-made context. They must write the question as well as verify the values chosen by finding the solution.

2.7.5 Wrap-Up: Cool Down

Component Overview: This Wrap-Up assesses students' place value and arithmetic reasoning in relation to rational numbers and operations. Students record whether they agree, disagree, or are unsure about the statements.



2.7.5 Wrap-Up: Cool Down

For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	<i>Answers will vary. Unsure, it depends on if the number is a whole number or a decimal number.</i>
When you add two positive decimals together, the value is always larger.	<i>Answers will vary. Agree, the decimal number will grow.</i>
To add fractions, their denominators must be the same.	<i>Answers will vary. Agree, to add fractions, they must first be like fractions.</i>
The product of two fractions is always a larger number.	<i>Answers will vary. Disagree, when you multiply fractions the product is smaller than the factors.</i>
When you put a zero on the end of any number, it makes it larger.	<i>Answers will vary. Unsure, it depends on whether the number is a whole number or a decimal number.</i>



Differentiating by Product:

- **Link Up:** Students write whether they agree or disagree on a whiteboard or use their hand to indicate. Students then must “Link Up” and find someone who does not have the same answer and try to convince them why their answer is correct.
- **A/S/N:** Students must include “Always,” “Sometimes,” or “Never” in their responses. The first and last prompts will be the “Sometimes” in which students must engage in mathematical discourse.
- **Prove It:** Challenge students to provide evidence for each answer with a justification. Correct answers will only be accepted if there is an example. Counterexamples included will be worth bonus points!